# Yue Samuel Lu

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## Research Interests

Galaxy formation/evolution, the circum/inter-galactic medium, cosmic rays, magnetic fields, large-scale structures, numerical simulations

## Education

Sep 2022 – Present
**University of California, San Diego (UCSD)**
Ph.D. in *Physics* (expected: 2027-2028)
Thesis: *Non-thermal processes in galaxy formation and evolution*
C.Phil./M.S. in *Physics* (Sep 2024)
GPA: 3.90/4.00
Advisor: *Dušan Kereš*

Sep 2018 – Jun 2022
**University of California, Santa Barbara (UCSB)**
B.S. in *Physics*; B.S. in *Mathematics*; Minor in *Astronomy*
Overall GPA: 3.82/4.00 (Physics GPA: 3.93, Math GPA: 3.92)
Consecutive Dean's Honors

## Research Positions

2022 – Present
**Graduate Student Researcher in FIRE Simulation Project** (UCSD)
Studied non-thermal processes in galaxy formation/evolution in FIRE simulations, focusing on cosmic-ray transport and magnetic fields

2020 – Present
**Student Researcher on Intergalactic Filaments** (KITP, UCSD)
Analyzed structure and dynamics of intergalactic filaments in cosmological simulations; connected to observables (Ly$\alpha$, X-ray); developed analytic models

2020 – 2022
**Undergraduate Researcher on AGN Accretion Disk** (UCSB)
Explored MHD simulations run by the `Athena++` code; analyzed an $m = 2$ anomaly in a simulated AGN accretion disk

## Selected Research Presentations

Sep 2026
**Symposium talk**, Galaxy Formation and Evolution in Southern California (GalFRESCA) 2026, UC Irvine

Aug 2026
**Conference talk**, 2026 Santa Cruz Galaxy Workshop, UC Santa Cruz

May 2026
**Invited seminar**, Center for Cosmology and AstroParticle Physics (CCAPP), OSU, Columbus, OH
**Conference talk**, Cosmic Ray Transport Workshop, CCAPP, OSU, Columbus, OH

Mar 2026
**Invited seminar**, UBC Physics and Astronomy Cosmos Coffee, Vancouver, BC, Canada

Dec 2025
**Informal talk**, Department of Astronomy, Tsinghua University, Beijing, China
**Informal talk**, Fudan Center for Astronomy and Astrophysics, Shanghai, China

Sep 2025
**Symposium talk**, GalFRESCA 2025, UC San Diego

Aug 2025
**Poster**, Cosmic Ecosystems, Perimeter Institute for Theoretical Physics, Waterloo, ON, Canada

Jun 2025
**Invited seminar**, Max Planck Institute for Extraterrestrial Physics, online

## Selected Research Presentations (continued)

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| Sep 2024 | 🔖 | **Symposium talk**, GalFRESCA 2024, Carnegie Observatories, Pasadena, CA |
| Aug 2024 | 🔖 | **Conference talk** (video), 2024 Santa Cruz Galaxy Workshop, UC Santa Cruz |
| Dec 2023 | 🔖 | **Poster**, International Conference on Resolving Galaxy Ecosystems on All Scales, CUHK, Hong Kong, China |
| Aug 2023 | 🔖 | **Conference talk** (video), 2023 Santa Cruz Galaxy Workshop, UC Santa Cruz |
| Sep 2021 | 🔖 | **Symposium talk** (video), UCSB Undergraduate Physics Research Symposium , online |

## Publications

### As the first author

1. **Y. S. Lu**, D. Kereš, P. F. Hopkins, and S. B. Ponnada, *Magnetic field evolution in feedback-dominated galaxies*, Oct. 2026. arXiv: 2610.17216 [astro-ph.GA].
2. **Y. S. Lu**, D. Kereš, P. F. Hopkins, S. B. Ponnada, C.-A. Faucher-Giguère, and C. B. Hummels, *Constraining cosmic ray transport models using circumgalactic medium properties and observables*, Jan. 2026. 🔗 DOI: 10.1093/mnras/staf1984. arXiv: 2505.13597 [astro-ph.GA].
3. **Y. S. Lu**, N. Mandelker, S. P. Oh, A. Dekel, F. C. van den Bosch, V. Springel, D. Nagai, and F. van de Voort, *The structure and dynamics of massive high-z cosmic-web filaments: three radial zones in filament cross-sections*, Feb. 2024. 🔗 DOI: 10.1093/mnras/stad3779. arXiv: 2306.03966 [astro-ph.CO].

### As a major contributor

1. S. B. Ponnada, P. F. Hopkins, **Y. S. Lu**, E. M. Silich, I. S. Butsky, and D. Kereš, *Strong Evidence for Cosmic-Ray-supported ∼L* Galaxy Halos via X-Ray and tSZ Constraints*, Jan. 2026. 🔗 DOI: 10.3847/2041-8213/ae2fd9. arXiv: 2510.13959 [astro-ph.GA].
2. M. Roy, K.-Y. Su, S. Tonnesen, **Y. S. Lu**, C. Hummels, and S. B. Ponnada, *To Survive or To Shatter: The Impact of Cosmic Rays on the Fate of Stripped Cold Clouds*, Jun. 2026. 🔗 DOI: 10.3847/1538-4357/ae664f. arXiv: 2510.21699 [astro-ph.GA].

### As a minor contributor

1. K.-Y. Su, G. L. Bryan, P. F. Hopkins, P. Natarajan, S. B. Ponnada, R. Emami, **Y. S. Lu**, and M. Roy, *Modelling cosmic rays at AGN jet-driven shock fronts*, Jan. 2026. 🔗 DOI: 10.1093/mnras/staf2060. arXiv: 2502.00927 [astro-ph.GA].
2. S. B. Ponnada, R. K. Cochrane, P. F. Hopkins, I. S. Butsky, S. Wellons, N. N. Sanchez, C. Hummels, **Y. S. Lu**, D. Kereš, and C. C. Hayward, *Hooks, Lines, and Sinkers: How Active Galactic Nucleus Feedback and Cosmic-Ray Transport Shape the Far-infrared–Radio Correlation of Galaxies*, Feb. 2025. 🔗 DOI: 10.3847/1538-4357/ada280. arXiv: 2410.02971 [astro-ph.GA].

## Selected Awards

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|---|---|---|
| 2025 | 🔖 | **Academic Senate Award**, UCSD |
| 2024 | 🔖 | **Distinguished Junior Teaching Award**, UCSD Physics department |
| 2023 | 🔖 | **Chair's Challenge Award**, UCSD Physics department |
| 2022 | 🔖 | **Graduation Honor**, UCSB College of Letters and Sciences<br>**Department Honor**, UCSB Physics Department |

## Teaching Experiences

Summer 2026 — **Instructor of Record for 1-series lab**, UCSD Physics Department
Administered the lower-division pre-health/biology-major physics lab; managed the LTACs and TAs; piloted on creating new evaluation formats for the course

Fall 2024–Present — **Lab TA Coordinator (LTAC) for 1-series lab**, UCSD Physics Department
Oversaw a $\sim$ 1,500-student pre-health/biology-major physics lab; ensured smooth course delivery; trained TAs/graders; managed student feedback

Fall 2022–Present — **Teaching Assistant**, UCSD Physics and A&A Department
Led discussion sections and lab sections for undergraduate level physics and astronomy courses; graded homework assignments and/or exams.
Courses taught:

- **PHYS 1-series lab:** introductory lab course designed mainly for pre-health students
- **PHYS 2-series:** general physics aimed for science/engineering majors
- **PHYS 7/ASTR 2:** galaxies and cosmology (general education level)
- **PHYS 13/ASTR 4:** life in the universe (general education level)
- **PHYS 163/ASTR 123:** galaxies (designed for upper division science majors)
- **ASTR 103:** dynamics of radiation and fluid (designed for upper division astronomy majors)
- **ASTR 104:** thermal astrophysics (designed for upper division astronomy majors)

2020–2021 — **Math, Physics, and Engineering Tutor**, UCSB Campus Learning Assistance Services (CLAS)
Taught lower division math and physics courses; ran group tutorials and drop-in sessions

2019–2022 — **Learning Assistant and Grader**, UCSB Physics Department
Assisted teaching assistants on running physics course discussion sessions; graded assignments and/or exams

## Academic service and outreach

2026–2027 — **Co-chair for ArXiv Hours Committee**, UCSD A&A Department

2023–2025 — **Colloquium and Journal Club Committee Member**, UCSD A&A Department

Apr 2024 — **Volunteer**, 2024 Continental U.S. Solar Eclipse Outreach Event

## Skills

Human Languages — English (advanced), Mandarin Chinese (native), Spanish (elementary)

Coding Languages — Python, C/C++, bash, Matlab, Mathematica

Misc. — Academic research, teaching, training, consultation, LaTeX typesetting